University
Mohammed VI
Polytechnic

College of
Computing

Deliverable : Lab 3

Data Management Course

UM6P College of Computing

Professor : Karima Echihabi **Program :** Computer Engineering
Session : Fall 2026

Team Information

Team Name	UNO
Member 1	Daniel Tchonang
Member 2	Amine Sira
Member 3	Zaid Zehari
Member 4	Mohammed Oumane
Repository Link	`https://github.com/unocomputing/unodatabase.git`

1 Introduction

In this lab we were tasked to express non trivial queries over RLMS using Relational Algebra

2 Implementation & Results

1. $\pi_{firstName,\ lastName}\left(Person \bowtie_{madebyPid\ =\ pid} Reserves\right)$
2. $$\begin{aligned}&\rho\left(labPeople,\ \pi_{pid}\left(Attached\right)\right)\\&\rho\left(PIPeople,\ \pi_{pid}\left(\sigma_{role="PI"}\left(Participate\right)\right)\right)\\&LabPeople \cup PIPeople\end{aligned}$$
3. $$\begin{aligned}&\rho\left(CCLab,\ \pi_{labId}\left(\sigma_{building="College\ of\ Computing"}\left(Laboratory\right)\right)\right)\\&\rho\left(Q,\ EquipmentModel \bowtie_{modelId} EquipmentUnit\right)\\&\rho\left(CPUServerLab,\ \sigma_{category="CPUServer"}\left(Q\right)\right)\\&CCLab \cup CPUServerLab\end{aligned}$$
4. $$\begin{aligned}&\rho\left(CPUEquip,\ \sigma_{category="CPU\ server"}\left(EquipmentModel \bowtie_{modelId} EquipmentUnit\right)\right)\\&\rho\left(GPUEquip,\ \sigma_{category="GPU\ Server"}\left(EquipmentModel \bowtie_{modelId} EquipmentUnit\right)\right)\\&\pi_{labId}\left(CPUEquip\right) \cup \pi_{labId}\left(GPUEquip\right)\end{aligned}$$
5. $$\begin{aligned}&\rho\left(CCLab,\ \pi_{labId}\left(\sigma_{building="College\ of\ Computing"}\left(Laboratory\right)\right)\right)\\&\rho\left(LabFaculty,\ \pi_{supervisorPid,\ labId}\left(Laboratory\right)\right)\\&\rho\left(FacultySupervisingCC,\ LabFaculty\ /\ CCLab\right)\\&\rho\left(FacultySupervisingCC(1 \rightarrow\ pid),\ FacultySupervisingCC\right)\\&Faculty \bowtie FacultySupervisingCC\end{aligned}$$
6. $$\begin{aligned}&\rho\left(BudgetFundLab2,\ \pi_{budgetLine}\left(\sigma_{labId=2}\left(FundsLab\right)\right)\right)\\&\rho\left(ProjectFunded,\ \pi_{projectCode}\left(FundsPrj \bowtie BudgetFundLab2\right)\right)\\&\pi_{pid}\left(Participates\ /\ ProjectFunded\right)\end{aligned}$$
7. $$\begin{aligned}&\rho\left(Cnt(1 \rightarrow pid, 2 \rightarrow n),\ {}_{madeByPid}\mathcal{G}_{\mathrm{COUNT}(resId)}(Reservation)\right)\\&\rho\left(C1(1 \rightarrow p1, 2 \rightarrow n1),\ Cnt\right)\\&\rho\left(C2(1 \rightarrow p2, 2 \rightarrow n2),\ Cnt\right)\\&\rho\left(NoRes(1 \rightarrow p2),\ \pi_{pid}\left(Person\right) - \pi_{pid}\left(Cnt\right)\right)\\&\rho\left(P1(1 \rightarrow p1),\ \pi_{pid}\left(Cnt\right)\right)\\&\pi_{p1,\ p2}\left(\sigma_{n1>n2}\left(C1 \times C2\right)\right) \cup \left(P1 \times NoRes\right)\end{aligned}$$

8.

$$\rho\,(EquipResByPeople,\ \pi_{pid,\ serialNo}\,(Reserves \bowtie Reservation))$$
$$\rho\,(Q,\ \pi_{pid,\ serialNo,\ labId}\,(EquipmentResByPeople \bowtie EquipmentUnit))$$
$$\rho\,(Pair(1{\to}pid1,\ 2{\to}s1,\ 3{\to}li1,\ 4{\to}pid2,\ 5{\to}s2,\ 6{\to}li2),\ Q \times Q)$$
$$\pi_{pid1}\,(\sigma_{(pid1=pid2)\wedge(s1>s2)land(li1>li2)}\,(Pair))$$

9. Let's set : TS1=timestamp(01/09/2026 00 :00 :00) and TS2=timestamp(01/10/2026 00 :00 :00)

$$\rho\,(UnitAIMov,\ \pi_{serialNo}\,(EquipmentUnit \bowtie \sigma_{name="AI\ Movement"}\,(Laboratory)))$$
$$\rho\,(ResDate,\ \pi_{resId}\,(\sigma_{(submissionTS \geq TS1)\wedge(submissionTS < TS2)}\,(Reservation)))$$
$$\pi_{resid}\,(ResDate \bowtie UnitAIMov)$$

10.

$$\rho\,(PIPeople,\ \pi_{pid,\ projectCode}\,(\sigma_{role="PI"}\,(Participates)))$$
$$\rho\,(PIPair(1 \to pid1, 2 \to pc1, 3 \to pid2, 4 \to pc2),\ PIPeople \times PIPeople)$$
$$\pi_{pid1}\,(\sigma_{(pid1=pid2)\wedge(pc1\neq pc2)}\,(PIPair))$$

11.

$$\rho\,(EquipResLab,\ \pi_{resId,LabId}\,(Reserves \bowtie EquipmentUnit))$$
$$\rho\,(Pair(1 \to rid1, 2 \to lid1, 3 \to rid2, 4 \to lid2),\ EquipResLab \times EquipResLab)$$
$$\rho\,(Q,\ \sigma_{(rid1=rid2)\wedge(lid1>lid2)}\,(Pair))$$
$$\rho\,(Q(1 \to rid),\ Q)$$
$$\pi_{approverPid}\,(Q \bowtie Reservation)$$

12. Let's set : TS1=timestamp(06/11/2026 00 :00 :00) and TS2=timestamp(07/11/2026 00 :00 :00)

$$\pi_{madeByPid}\,(\sigma_{(plannedStart\geq TS2)\vee(plannedEnd<TS1)}\,(Reservation))$$

13.

$$\rho\,(LC(1 \to labId,\ 2 \to nb),\ {}_{labId}\mathcal{G}_{\mathrm{COUNT}(serialNo)}\,(EquipmentUnit))$$
$$\rho\,(EmptyLab,\ \pi_{labId}\,(Laboratory) - \pi_{labId}\,(LC))$$
$$\rho\,(LabCnt,\ LC \cup (EmptyLab \times \rho\,(Z(1 \to nb),\ \{(0)\})))$$
$$\rho\,(Avg(1 \to moy),\ \mathcal{G}_{\mathrm{AVG}(nb)}\,(LabCnt))$$
$$\pi_{labId}\,(\sigma_{nb<moy}\,(LabCnt \times Avg))$$

14.

$$\rho\,(RL,\ \pi_{resId,\ labId}\,(Reserves \bowtie EquipmentUnit))$$
$$\rho\,(AppRL,\ \pi_{madeByPid,\ labId,\ resId}\,(\sigma_{status=\text{'Approved'}}\,(Reservation) \bowtie RL))$$
$$\rho\,(Cnt(1 \to labId,\ 2 \to pid,\ 3 \to nb),\ {}_{labId,\ madeByPid}\mathcal{G}_{\mathrm{COUNT}(resId)}\,(AppRL))$$
$$\rho\,(Mx(1 \to labId,\ 2 \to mx),\ {}_{labId}\mathcal{G}_{\mathrm{MAX}(nb)}\,(Cnt))$$
$$\pi_{labId,\ pid}\,(\sigma_{nb=mx}\,(Cnt \bowtie Mx))$$

15. Let's set : TS1=timestamp(01/01/2025 00 :00 :00) and TS2=timestamp(01/01/2026 00 :00 :00)

$\rho\left(PRD,\ \pi_{resId,madeByPid}\left(\sigma_{(submissionTS\geq TS1)\wedge(submissionTS<TS2)}\left(Reservation\right)\right)\right)$
$\rho\left(Triplet(1\rightarrow ri1, 2\rightarrow pi1, 3\rightarrow ri2, 4\rightarrow pi2, 5\rightarrow ri3, 6\rightarrow pi3),\ PRD\times PRD\times PRD\right)$
$\pi_{pid1}\left(\sigma_{(pid1=pid2)\wedge(pid2=pid3)\wedge(ri1>ri2)\wedge(ri2>ri3)}\left(Triplet\right)\right)$

3 Challenges

We did face a few challenges including :
— Renaming a few columns to take into account the ones with identical names when we join or when we use the cross product.
— Tracking the attributes given by every command.